Determining the Charge to Mass Ratio of an Electron

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1 Introduction

The purpose of this experiment is to determine the charge-to-mass ratio of the electron by observing the motion of an electron beam in a magnetic field. When electrons move through a magnetic field perpendicular to their velocity, they experience a Lorentz force that causes them to travel in a circular path [1]. By measuring the radius of this path for known accelerating voltages in a roughly constant magnetic field created by a Helmholtz coil, the value of the ratio of the charge of an electron e and the mass of an electron m, $\frac{e}{m}$ was determined to be $\frac{e}{m} = (3.93 \pm 0.04) \cdot 10^{11}$ C kg^{-1}, which is in disagreement with the accepted value of $1.58 \cdot 10^{11}$ C kg^{-1} [2]. Reasons for this discrepancy include a difficulty in obtaining an accurate measurement of the circular electron path radius, as well as potential non uniformities in the off centre parts of the Helmholtz coil.

2 Equipment and Methodology

2.1 Equipment

The experimental setup, as seen in figure 1, consisted of a glass bulb containing an electron gun placed between a pair of Helmholtz coils, with an external diameter of 31 ± 0.2 cm and a centre-to-centre distance of 17 ± 0.2 cm. The apparatus also included two power supplies, a rheostat, an ammeter, and a voltmeter. We chose to use an uncertainty for the ammeter and voltmeter that was much greater than the stated values. The last few digits in the current and voltage values were fluctuating, so we decided to take the values to 3 decimal places for current and 2 decimal places for voltage. This gives us an uncertainty of ± 0.005 V for the voltmeter and ± 0.0005 A for the ammeter. An illuminated scale and plastic reflector were used to measure the diameter of the circular electron trajectory. The scale projected an image of a ruler, which was positioned such that the camera capturing the image (not seen in figure) had the beam and image of the ruler in the same plane. The relative error in placement was assumed to be negligible compared to the uncertainty in the actual measurement. The ruler was not labelled with numbers but it was measured that individual tick marks were 1 mm; the ruler did not have tick marks in the middle which can be seen in figure 2, so the middle was not used for edge measurements, but was determined to be 2 cm wide. Measurements were taken to be accurate within ± 1 mm, as that is the sum of the uncertainties on both ends.

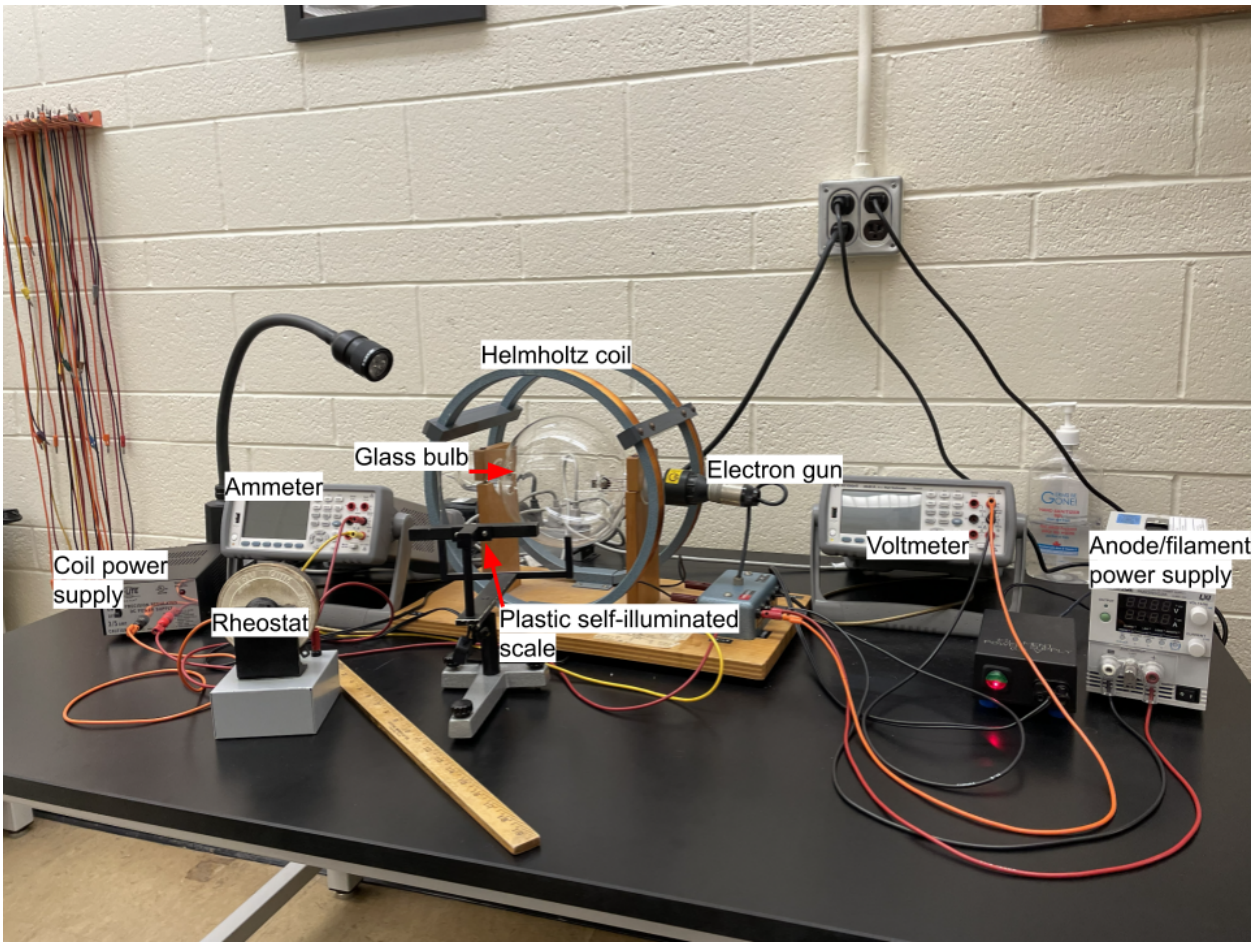

Figure 1: Our labelled experimental setup. In actual experiments, the plastic scale was placed roughly in between our camera and the electron beam, such that the image of the scale was on the beam. The positions of the electric devices remained constant during the series of experiments to minimize electromagnetic interference on the beam.

2.2 Methodology

The filament power supply was first turned on and allowed to heat for approximately 30 seconds to ensure sufficient emission from the cathode. The accelerating voltage was then applied to the electron gun, producing a visible electron beam inside the glass bulb. Current was then applied to the Helmholtz coils to generate a magnetic field perpendicular to the path of the electrons. The glass bulb was rotated slightly until the electron beam formed a roughly circular path that did not touch the glass bulb. The diameter of the circular electron orbit was measured by taking pictures of the illuminated scale positioned in the plane of the beam. We first varied the current in the coils while keeping the accelerating voltage constant. We then varied the accelerating voltage while maintaining a constant coil current. Lastly, we put a phone at varying distances and varying coil currents to qualitatively measure the effects of an external electromagnetic field on the electron beam.

3 Results and Discussion

3.1 Anomalous behaviours at low voltage and high current

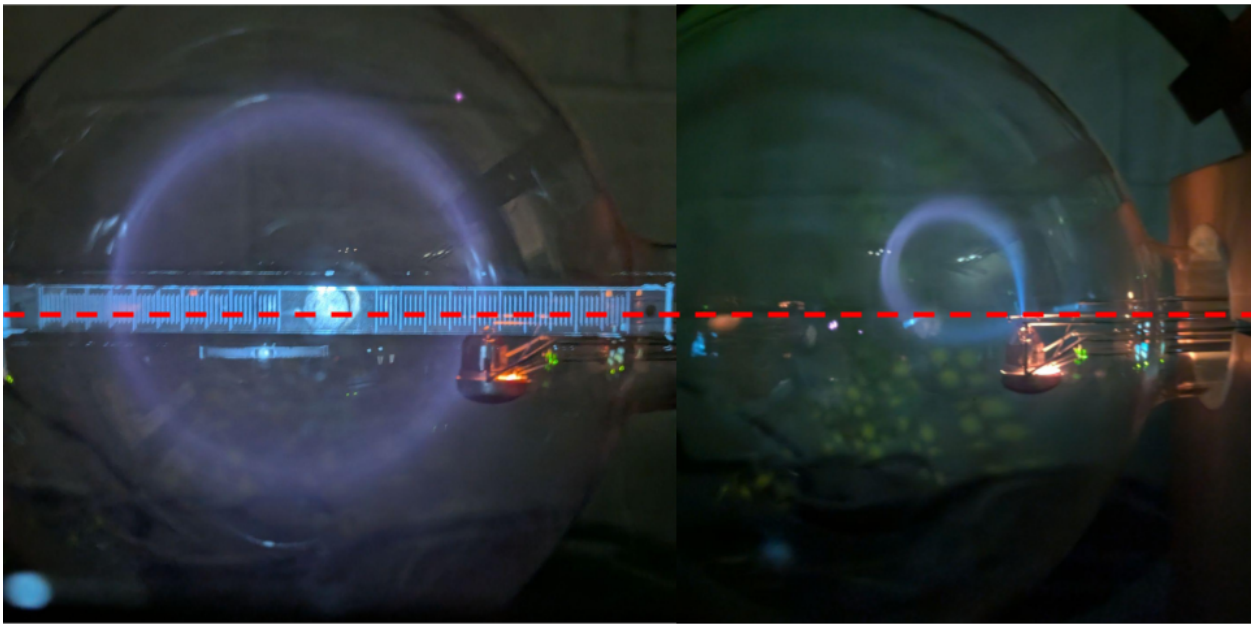

Figure 2: Comparing the expected electron beam behaviour on the left with the anomalous behaviour on the right. We mark both images with a dotted red line to mark the approximate centre of the bulb. We can see that in normal behaviour, the centre of the bulb roughly lines up with the centre of the beam. In the anomalous case, the beam rests almost entirely above the centre line. Additionally, we can see the non-marked middle of the ruler, which we determined to be 2 cm wide.

The anomalous behaviour observed at low accelerating voltage and high coil current can be explained by the increased sensitivity of the electron beam to small changes in the magnetic field. At the lower accelerating voltage, the electrons in the electron beam travel more slowly, which means that the Lorentz force produced by the magnetic field is smaller making the beam radius smaller and more sensitive to imperfections in the field [3]. As mentioned, the Helmholtz coils do not produce a perfectly uniform field, particularly away from the central axis. Any transverse field components or spatial non-uniformities then cause the orbit to deviate from a centred circular path, producing the upward-shifted and distorted trajectory seen on the right side of Figure 2. This effect is amplified at small radii, since for small radii, a perturbation in the magnetic field produces a larger fractional change in the radius and therefore centre of the orbit.

To mitigate these effects, we always took measurements with a relatively large radius, such that the distance between the center of the current loop and the center of the Helmholtz coil was less than 3 cm, so $\rho < 0.2R$. As such, we introduce a correction factor of $(1 - \frac{0.075}{100})$ in all of our calculations for the field generated by the Helmholtz coils, as can be seen in equations 2, 3 and 6.

3.2 Determining the Value of B_e

$$B_c = \sqrt{\frac{2m}{e}\Delta V}\frac{1}{r} - B_e \tag{1}$$

In conjunction with the equation determining B_c computationally:

$$B_c = (\frac{4}{5})^{3/2}\frac{\mu_0 nI}{R}(1 - \frac{0.075}{100}) \tag{2}$$

We can combine the two equations above, yielding a relationship between $\frac{1}{r}$ and the magnetic field generated by the Helmholtz coils (on the left side of the equation):

$$(\frac{4}{5})^{3/2}\frac{\mu_0 nI}{R}(1-\frac{0.075}{100}) = \sqrt{\frac{2m}{e}\Delta V}\frac{1}{r} - B_e \tag{3}$$

Thus, the external magnetic field B_e is the negative y-intercept of the linear relationship between the inverse relationship and the calculated field generated by the Helmholtz coils. Using our radius measurements taken at varying currents and constant voltage difference (seen below), we can plot the data and determine B_e to be $-(1.3 \pm 0.6) * 10^{-4}$ T.

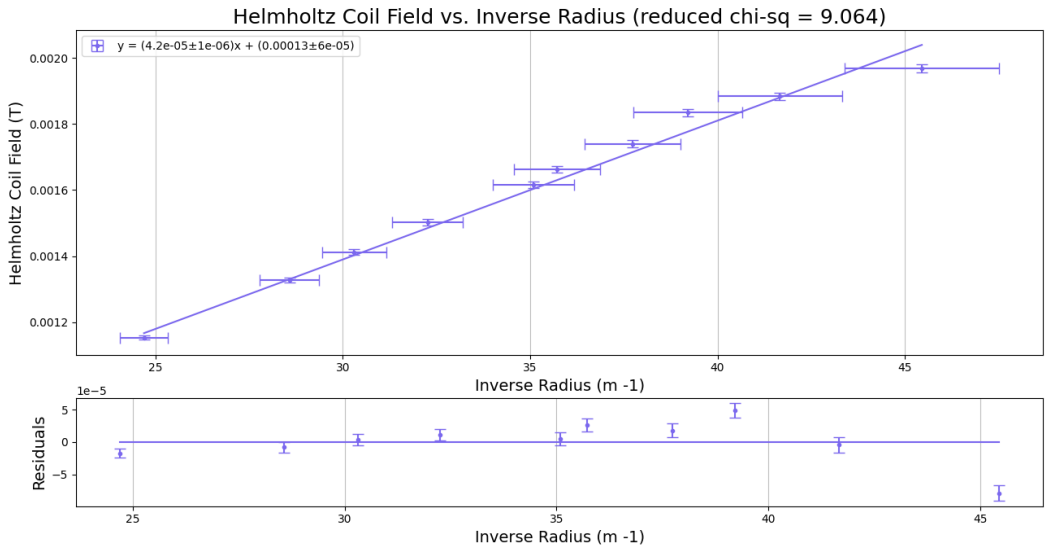

Figure 3: A graph showing the relationship between the inverse radius (in m^{-1}) and the magnetic field generated by the Helmholtz coils (in T). The reduced chi-squared value is 9.064, implying potential underfitting.

Our calculated value for B_e does not agree with the reference value for the magnetic field present in Toronto, which is around $5.2974 * 10^{-5}$ T [4]. However, this disparity can be explained by the presence of other sources of magnetic field, like the other Helmholtz coils present in the room, and the relatively large reduced χ^2 value of 9.064 implying a potential underfitting.

3.3 Determining the Charge to Mass Ratio of an electron

To determine the Charge to Mass Ratio of an electron, we use the equation:

$$\frac{1}{r} = \sqrt{\frac{e}{m}}k\frac{I_T}{\sqrt{\Delta V}} \tag{4}$$

Where k and I_T are determined using the formulae:

$$k = \frac{1}{\sqrt{2}}(\frac{4}{5})^{\frac{3}{2}}\frac{\mu_0 n}{R} \tag{5}$$

$$I_T = I(1-\frac{0.075}{100}) + \frac{B_e}{\sqrt{2}k} \tag{6}$$

We can plot the relationship between the inverse radius and the value $\frac{I_T}{\sqrt{\Delta V}}$, which should yield a proportional relationship with a slope of $\sqrt{\frac{e}{m}}k$.

When collecting data to plot this relationship, we tried our best to maintain a constant current through

the Helmholtz coils (I), though there were still some slight variations in current, which motivated the use of a relationship between $\frac{I_T}{\sqrt{\Delta V}}$ and $\frac{1}{r}$ instead of a relationship between $\frac{1}{\sqrt{\Delta V}}$ and $\frac{1}{r}$.

Our data, and the resulting proportional relationship, can be seen plotted below:

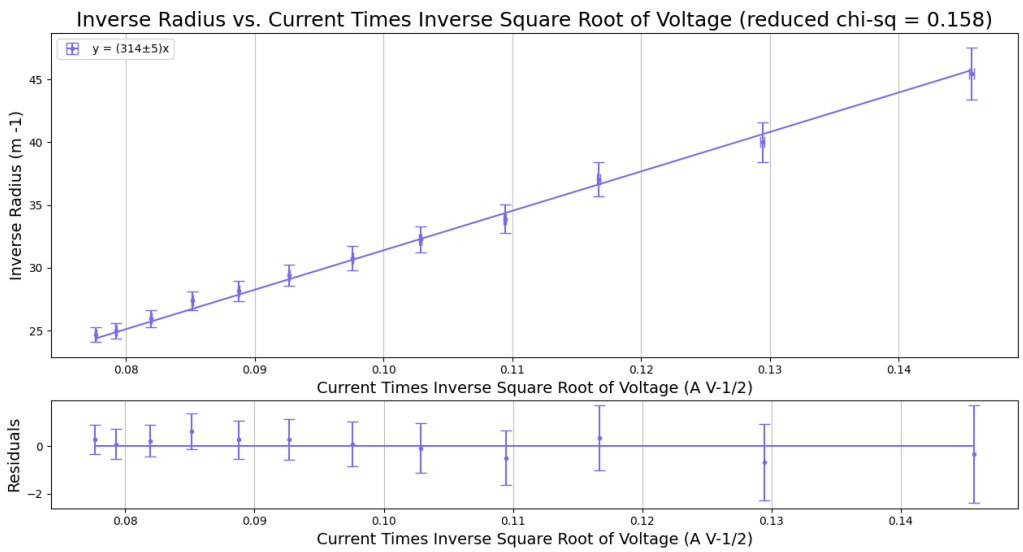

Figure 4: A graph showing the relationship between the product of the total current and the inverse square root of the voltage (in A $V^{-\frac{1}{2}}$, and the inverse radius (in m^{-1}). The reduced chi-squared value is 0.158, implying potential overfitting.

As per equation 4, we can determine the value of $\frac{e}{m}$ from the slope by dividing by k and squaring the results. Using the given values of $n = 130$ and $R = 0.165 \pm 0.001$, the calculated value of k is approximately:

$$k = (5.01 \pm 0.03) \cdot 10^{-4} \tag{7}$$

We can then substitute the given value of k to determine the charge to mass ratio:

$$\frac{e}{m} = (\frac{314}{k})^2$$

$$\frac{e}{m} = (\frac{314}{5.01 \cdot 10^{-4}})^2$$

$$\frac{e}{m} = (3.93 \pm 0.09) \cdot 10^{11} \text{ C kg}^{-1} \tag{8}$$

This value does not entirely agree with the reference value of the charge to mass ratio of an electron, approximately $1.58 \cdot 10^{11}$ C kg^{-1} [2], even within the bounds of our uncertainty. This disparity may have been because of overfitting, as our reduced χ^2 value of 0.158 implies, and because of the non-uniformity of the field generated by the Helmholtz coils, which the formula we used did not consider.

4 Uncertainty Analysis

4.1 Uncertainty Propagation for Constants

All of the constants in this lab were products of our uncertain values, so all uncertainty propagation necessary for these constants can be represented by the formula [5]:

$$\Delta z = z\sqrt{\sum_{i=0}^{n}(\frac{\Delta x_i}{x_i})^2} \tag{9}$$

Where z is a product or quotient of $x_1, x_2, \cdots, x_n$. The uncertainties are represented with the terms Δz and Δx_i.

4.2 Uncertainty Propagation for Linear and Proportional Fits

For a linear fit given by the equation $y = mx + b$ (in the case of a proportional fit, $b = 0$), the uncertainties for the slope m and y-intercept b are given by the equations [5]:

$$s_m = \sqrt{N\frac{s_{y,x}^2}{\Delta}} = \sqrt{\frac{N}{N-2}\frac{\sum[y_i - (mx_i + b)]^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{10}$$

$$s_b = \sqrt{\frac{s_{y,x}^2 \sum x_i^2}{\Delta}} = \sqrt{\frac{1}{(N-2)}\frac{\sum[y_i - (mx_i + b)]^2 \sum x_i^2}{N\sum x_i^2 - (\sum x_i)^2}} \tag{11}$$

$$\Delta = N\sum x_i^2 - (\sum x_i)^2 \tag{12}$$

$$s_{y,x}^2 = \frac{1}{N-2}\sum[y_i - (mx_i + b)]^2 \tag{13}$$

4.3 Sample Calculations

4.3.1 Sample Calculations for Uncertainty of k

For this lab, k was calculated using the equation:

$$k = \frac{1}{\sqrt{2}}(\frac{4}{5})^{\frac{3}{2}}\frac{\mu_0 n}{R} \tag{14}$$

With $n = 130$ and $R = 0.165$, with an uncertainty $\Delta R = 0.001$. The resulting uncertainty in k can be calculated quite simply:

$$\Delta k = k\sqrt{(\frac{\Delta R}{R})^2}$$

$$\Delta k = (5.01 \cdot 10^{-4})\sqrt{(\frac{0.001}{0.165})^2}$$

$$\Delta k = 3 \cdot 10^{-6}$$

4.3.2 Sample Calculations for Uncertainty in Slope and Y-Intercept

We perform sample calculations for the uncertainty in slope and y-intercept for the relationship between the field generated by the Helmholtz coils and the inverse radius. The data for these calculations can be found in Appendix A.

$$s_m = \sqrt{\frac{10}{8}\frac{[(0.00197 - (4.2 \cdot 10^{-5} \cdot 45.5 + 0.00013)]^2 + \cdots + [(0.00115 - (4.2 \cdot 10^{-5} \cdot 24.7 + 0.00013)]^2}{8(45.5^2 + 41.7^2 + \cdots + 24.7^2) - (45.5 + 41.7 + \cdots + 24.7)^2}}$$

$$s_m = 1 \cdot 10^{-6}$$

$$s_b = \sqrt{\frac{1}{(8)}\frac{([(0.00197 - (4.2 \cdot 10^{-5} \cdot 45.5 + 0.00013)]^2 + \cdots [0.00115 \cdots]]^2(54.5^2 + 41.7^2 + \cdots + 24.7^2)}{8(45.5^2 + 41.7^2 + \cdots + 24.7^2) - (45.5 + 41.7 + \cdots + 24.7)^2}}$$

$$s_b = 6 \cdot 10^{-5}$$

4.4 Sources of error and uncertainty

The largest uncertainty arose from measuring the diameter of the electron beam. The beam had a finite thickness, making it difficult to determine the exact position of its edge. The uncertainties in the radius measurement significantly affected the calculated value of the charge-mass ratio, as the radius is squared. To minimize this, we used the self-illuminated scale was placed so that its image, seen directly and in the plastic reflector, lined up with the electron beam in the same apparent plane. When this condition was reached, parallax error was minimized because the measurement was effectively being made along a line of sight perpendicular to the beam.

External magnetic fields from nearby equipment or ferromagnetic materials in the laboratory may have slightly altered the effective magnetic field. We aimed to keep most electromagnetic sources constant (i.e. the distances of our phones and laptops from the coil), however the field would have had some variation, particularly due to the existence of other people and their associated electronic devices that moved around the room during our experiments. We did note that the effect of a phone only became noticeable when it was placed at very close distances to the bulb (within roughly 10 cm), as that is when we started to see distortion in the circular path of the beam. We note that this behaviour was most prevalent at lower currents, and that the beam was typically drawn towards the direction of the phone.

References

[1] H. Zhan, E. Horsley, and A. Harlick. Charge-to-mass ratio for the electron, 2025. Undergraduate physics laboratory manual.

[2] National Institute of Standards and Technology (NIST). Electron charge to mass ratio, 2026. Accessed: 2026-03-10.

[3] David J. Griffiths. *Introduction to Electrodynamics*. Cambridge University Press, Cambridge, 4 edition, 2017.

[4] Magnetic-Declination.com. Magnetic field in toronto, canada, 2026. Accessed: 2026-03-10.

[5] David M. Harrison. *Error Analysis in Experimental Physical Science*. Department of Physics, University of Toronto, 2011. Last revision: September 10, 2011.

5 Appendix: Data

Note that data here is reported to the significant figures generated by the spreadsheet for completeness, we truncated it prior to processing as mentioned in our actual report.

5.1 Inverse radius vs current scaled by $V^{-1/2}$

Current $\times V^{-1/2}$ (A V$^{-1/2}$)	Inverse Radius (m^{-1})	X_{err}	Y_{err}
0.1456783849	45.45454545	0.0001847425527	2.066115702
0.1294234885	40	0.0001382904703	1.6
0.1166902677	37.03703704	0.0001102052871	1.371742112
0.1094239017	33.89830508	0.00009629319255	1.149095088
0.1028752915	32.25806452	0.00008516752403	1.040582726
0.09759858773	30.76923077	0.00007744591279	0.9467455621
0.09265621251	29.41176471	0.00007060875311	0.8650519031
0.08876817422	28.16901408	0.00006583781306	0.7934933545
0.08515245273	27.39726027	0.00006170164463	0.7506098705
0.08192620872	25.97402597	0.00005824090877	0.6746500253
0.07925650137	25	0.00005535713654	0.625
0.07765652258	24.69135802	0.00005379277861	0.6096631611

5.2 Inverse radius vs field strength (T)

Inverse Radius (m^{-1})	Helmholtz Coil Field (T)	X_{err}	Y_{err}
45.45454545	0.001968699352	2.066115702	0.001
41.66666667	0.001883750081	1.666666667	0.001
39.21568627	0.00183419634	1.452432825	0.001
37.73584906	0.001740044232	1.279181324	0.001
35.71428571	0.001662881977	1.152073733	0.001
35.0877193	0.001615451968	1.079622132	0.001
32.25806452	0.001502186273	0.9487666034	0.001
30.3030303	0.001411573718	0.8536064874	0.001
28.57142857	0.001327332357	0.782778865	0.001
24.69135802	0.001153894263	0.6413339747	0.001